

On the Integer Equation $6 + x^3 - y^2 + y^2 z^2 = 0$

Problem

Determine the integer solutions of

$$6 + x^3 - y^2 + y^2 z^2 = 0.$$

Equivalently,

$$x^3 + y^2(z^2 - 1) = -6 \quad \text{or} \quad y^2(1 - z^2) = x^3 + 6.$$

Step 1: Immediate special cases

Proposition 1. *If $y = 0$, there is no integer solution.*

Proof. Then $x^3 = -6$, impossible in $\mathbb{Z}$. □

Proposition 2. *If $z = \pm 1$, there is no integer solution.*

Proof. If $z^2 = 1$, the equation becomes again $x^3 = -6$, impossible. □

Step 2: Parity and modular restrictions

Proposition 3. *z cannot be odd.*

Proof. Assume z odd. Then $z^2 \equiv 1 \pmod{8}$, hence $z^2 - 1 \equiv 0 \pmod{8}$. Reducing

$$x^3 + y^2(z^2 - 1) = -6$$

modulo 8 gives $x^3 \equiv -6 \equiv 2 \pmod{8}$. But cube residues mod 8 are only 0, 1, 3, 5, 7. Contradiction. □

Proposition 4. *If a solution exists, then z is even, y is odd, and $x \equiv 3 \pmod{4}$.*

Proof. By the previous proposition, z is even. Then $z^2 \equiv 0 \pmod{4}$, so $z^2 - 1 \equiv 3 \pmod{4}$. Reduce

$$x^3 + y^2(z^2 - 1) = -6$$

modulo 4:

$$x^3 + 3y^2 \equiv 2 \pmod{4}.$$

If y were even, then $3y^2 \equiv 0 \pmod{4}$, so $x^3 \equiv 2 \pmod{4}$, impossible. Hence y is odd, so $y^2 \equiv 1 \pmod{4}$, giving $x^3 \equiv 3 \pmod{4}$, thus $x \equiv 3 \pmod{4}$. □

Step 3: The case $z = 0$

Proposition 5. *There is no integer solution with $z = 0$.*

Proof. If $z = 0$, the equation is

$$y^2 = x^3 + 6.$$

This is the Mordell curve $E : y^2 = x^3 + 6$.

To determine all integral points on E , one may use the standard elliptic-curve integral-point algorithm (Baker bounds + elliptic logarithms), implemented in SageMath:

```
EllipticCurve([0,0,0,0,6]).integral_points().
```

Running this computation returns the empty set:

$$E(\mathbb{Z}) = \emptyset.$$

Hence no integer pair (x, y) satisfies $y^2 = x^3 + 6$, so no solution exists with $z = 0$. $\square$

What is fully proved, and what remains

All of the following are now proved:

- no solutions with $y = 0$;
- no solutions with $z = \pm 1$;
- no solutions with odd z ;
- no solutions with $z = 0$.

Therefore any possible solution would have to satisfy

$$z \in 2\mathbb{Z} \setminus \{0\}, \quad y \equiv 1 \pmod{2}, \quad x \equiv 3 \pmod{4}.$$

Remark 1. *At this stage, a complete elimination of the remaining branch $z \in 2\mathbb{Z} \setminus \{0\}$ has not been established in this write-up. So this document gives a fully rigorous reduction plus complete resolution of all listed special cases, but not yet a full global classification.*

Sanity checks

(A) Substitution checks for all candidate special cases

- $y = 0$ forces $x^3 = -6$ (not an integer cube), so no candidate survives.
- $z = \pm 1$ forces $x^3 = -6$, so no candidate survives.
- $z = 0$ reduces to $y^2 = x^3 + 6$, and the integral-point computation gives no candidate.

(B) Wide bounded search (evidence only)

A direct search over

$$|x| \leq 200, \quad |y| \leq 200, \quad |z| \leq 50$$

finds no solution. Extending to

$$|y| \leq 3000, \quad |z| \leq 3000$$

with exact-cube checking for $x^3 = -6 - y^2(z^2 - 1)$ also finds none. This supports nonexistence but is not a proof for all integers.

Concise theorem statement and summary

Theorem 1 (Proved nonexistence in all listed special branches). *For*

$$6 + x^3 - y^2 + y^2 z^2 = 0, \quad x, y, z \in \mathbb{Z},$$

there are no solutions in each of the branches

$$y = 0, \quad z = \pm 1, \quad z \text{ odd}, \quad z = 0.$$

Hence any hypothetical solution must satisfy

$$z \in 2\mathbb{Z} \setminus \{0\}, \quad y \text{ odd}, \quad x \equiv 3 \pmod{4}.$$

Proof summary. Rewrite as $x^3 + y^2(z^2 - 1) = -6$. Immediate substitutions eliminate $y = 0$ and $z = \pm 1$. Modulo 8, odd z is impossible since it would force $x^3 \equiv 2 \pmod{8}$. For even z , modulo 4 forces y odd and $x \equiv 3 \pmod{4}$. Finally, $z = 0$ gives the Mordell curve $y^2 = x^3 + 6$, whose integral points are empty. Therefore all these branches are impossible.